

# NIRNAY PRATAP SINGH

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## SUMMARY

Computer Science undergraduate (B.Tech, 3rd year) focused on cloud infrastructure and applied GenAI, with a full-stack engineering foundation spanning database design, REST APIs, and production deployment. Has shipped RAG pipelines, vector search, and multi-tenant PostgreSQL systems on AWS and Vercel. Comfortable owning a feature end to end and turning ambiguous requirements into reliable, secure software.

## EDUCATION

### B.Tech in Computer Science Engineering | Bennett University

Aug 2024 – May 2028

3rd Year, 5th Semester | CGPA: 7.64 / 10 | Cloud Computing Course Grade: 9 / 10

Relevant Coursework: Cloud Computing, Data Structures & Algorithms, Design & Analysis of Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming.

## EXPERIENCE

### Data Analyst Intern | Lysandra Group

Jun 2025 – Present

- Cleaned and queried 50,000+ row business datasets weekly, surfacing insights that informed stakeholder decisions on resource allocation and campaign spend.
- Built and maintained 5+ SQL dashboards tracking sales and operations KPIs, reducing manual reporting effort by around 40% and freeing roughly 6 hours a week for deeper analysis.
- Automated recurring data-cleaning workflows in Python (pandas), cutting multi-hour processes down to minutes with no manual steps.
- Partnered with 3 cross-functional teams, translating vague data requests into structured, reusable weekly reports used in leadership reviews.

### Management Head | Alt Reality Club, Bennett University

Aug 2024 – May 2025

- Led a 12-person core team to deliver 8+ technical events and workshops, growing club membership 35% to 150+ active students over the academic year.

## PROJECTS

### VeriPolicy: AI Policy Intelligence Platform [Code](#) [Live](#)

React, TypeScript, PostgreSQL, Supabase, RAG, Llama 3.3, Vector Embeddings, Vercel — Jul 2026

- Built and shipped a full-stack platform (frontend, service layer, relational database) that indexes 2M+ records across 47 jurisdictions to produce referenced foresight briefs in under two seconds.
- Designed normalized PostgreSQL schemas with Row-Level Security for multi-tenant isolation and added vector search over embeddings to surface the closest historical analogues.
- Engineered a RAG pipeline on Llama 3.3-70B that grounds every AI claim in primary-source SIPRI and OWID data, fed by a live NewsData ingestion pipeline and deployed on Vercel with environment-based secrets.

### OpenEnv Code Review Agent [Code](#) [Live](#)

Python, FastAPI, Docker, Reinforcement Learning, Meta PyTorch OpenEnv — Mar–Apr 2026 (Meta PyTorch OpenEnv Hackathon × SST)

- Designed a 3-tier benchmark (easy to hard) covering bug detection, logic errors, and critical SQL injection vulnerabilities, with partial-credit reward shaping to mirror real code-review complexity.
- A first-pass Llama 3.3-70B baseline agent scored 93.3% on the benchmark, validating the environment design.
- Packaged everything in Docker behind a clean REST API (/reset, /step, /state) so any agent can plug in with minimal integration overhead.

### Air Sentinel AI: Air Quality Monitor [Code](#)

React, Python, scikit-learn, MongoDB, REST APIs — Dec 2025

- Trained a regression model on 2+ years of historical AQI data (93% accuracy on held-out test sets), running inference on sensor readings every 5 minutes.
- Built a self-contained system handling data ingestion, ML inference, and a live React dashboard in one deployable package; MongoDB backend handles 10,000+ time-series records with optimized indexing.

### Elevare: Full-Stack Mental Health Platform [Code](#) [Live](#)

React, Node.js, Express.js, MongoDB, Stripe, JWT — Dec 2024

- Built 12+ REST endpoints with JWT authentication and role-based access control across 4 user types, backed by well-structured MongoDB schemas.
- Added Stripe subscription billing to a mobile-first, responsive full-stack app supporting peer support and professional booking.

## TECHNICAL SKILLS

### Cloud & DevOps: AWS (EC2, S3, IAM, Lambda), Docker, Git/GitHub, Vercel, CI/CD

AI & GenAI: RAG, vector embeddings, LLM integration (Llama 3.3), pandas, scikit-learn, SQL analytics

Languages: C++, Java, Python, SQL, JavaScript (ES6+), TypeScript

Backend & APIs: Node.js, Express.js, FastAPI, JWT auth | Databases: PostgreSQL, MySQL, MongoDB

Frontend: React.js, TypeScript, Tailwind CSS, responsive design | CS Fundamentals: DSA, OOP, DBMS, OS, Computer Networks, SDLC

## ACHIEVEMENTS & CERTIFICATIONS

- Smart India Hackathon: qualified for and competed in the national rounds (2024 and 2025).
- Meta PyTorch OpenEnv Hackathon × SST 2026: RL-based code-review environment; baseline agent scored 93.3%.
- AWS Academy Graduate, Cloud Foundations • AWS Academy Graduate, Cloud Security Foundations
- Anthropic AI Fluency: Framework and Foundations • Machine Learning Capstone (Coursera)